Frequency Bin	Recovery (%)
High	30.576
Mid	19.794
Low	10.231

Table 7: **Pilot Frequency-Stratified Recovery (Zero-shot).** Exact recovery rates (%) across high-, mid-, and low-frequency bins.

sitivity to wording differences. All evaluation metrics aggregate over multiple references to avoid dependence on any single phrasing.

We evaluate the main reconstruction accuracy metric alongside semantic and logical metrics. While some items may appear in pretraining corpora, reconstruction accuracy is not interpreted in isolation. Instead, differences across semantic, logical, and robustness metrics are used to assess structure-aware semantic grounding beyond surface recall.

B.4 Frequency-Stratified Recovery Pilot

A potential concern is that exact recovery on four-character expressions may be influenced by memorization, since frequent idioms are more likely to appear in pretraining corpora. To quantify this effect, we conduct a pilot frequency-stratified analysis under the zero-shot setting using two representative models that span a wide capability range: GPT-4o and Qwen3-8B. We stratify expressions into three frequency bins (high, mid, and low) and measure the recovery rate as exact canonical reconstruction. Table 7 reports bin-wise recovery averaged across the two models.

As shown in Table 7, we observe a clear monotonic frequency effect: recovery is highest on high-frequency expressions, decreases on mid-frequency expressions, and remains non-trivial on low-frequency expressions. This indicates that lexical frequency contributes to reconstruction performance, while meaningful recovery persists even in the low-frequency regime.

C Prompt Templates for Idiom Meaning Inference

All baseline runs use a fixed two-message prompting scheme consisting of a system instruction and a user query template. The system instruction constrains the output format to a single-line response in Chinese. The user template provides the target idiom and repeats the required output schema to reduce formatting deviations. Across different experimental conditions, only the prompt text is modified, while the model, decoding settings, batch processing logic, and output parsing rules remain unchanged. For multilingual settings, the same prompting structure is retained, with the prompt content adapted through direct translation or equivalent reformulation while preserving identical functional roles and formatting requirements.

To facilitate reproducibility, Table 8 presents the baseline prompt templates used for idiom meaning inference across languages. The prompts are shown in their actual formatted layout, including line breaks, to illustrate the structural constraints imposed on model outputs. No task-specific examples, demonstrations, or additional guidance are included beyond the required format specification.

All experiments are conducted using fixed inference settings. For the *baseline* condition, no task-specific examples or chain-of-thought instructions are provided beyond the one-line format constraint. For the *CoT* condition, we use a fixed two-line CoT template (Table 10) while keeping all decoding settings identical.

D Semantic Hallucination vs. Idiomatic Retrieval

Figure 7 presents a representative failure case on the idiom *Qing Guo Qing Cheng*. Although the scrambled characters preserve strong lexical cues, some models still produce fluent but incorrect literal interpretations, reflecting compositional hallucination. In contrast, stronger models correctly retrieve the idiomatic metaphor, demonstrating that semantic plausibility and structural reconstruction can diverge sharply.

E Qualitative Analysis and Case Studies

In this section, we provide a qualitative visual analysis of model performance across different experimental settings. Figures 8 and 9 illustrate baseline (zero-shot) behaviors and highlight common failure modes such as hallucination and structural blindness. Figures 10 through 13 further compare Chain-of-Thought (CoT) and few-shot strategies for each language (ZH-CN, ZH-TW, JP, KO). These visualizations display the system prompts, user queries, and model outputs, and they illustrate the strict constraints imposed by our evaluation pipeline.

F Reference Performance on Canonical Inputs

To isolate the specific impact of structural perturbation, we conducted a control experiment using the canonical (unperturbed) forms of the expressions. This serves two critical purposes: establishing a performance upper bound for each model and validating the reliability of our automated diagnostic metrics.

Table 11 reports the diagnostic scores under this ideal condition. We observe that all evaluated models demonstrate high proficiency when the input structure is intact. For instance, semantic fidelity scores (S_{Acc}^{mean}) consistently range between 0.75 and 0.88 across different model tiers, and logical validity (S_{Logic}) remains high.

These results support two key conclusions regarding the main experiments:

1. **Knowledge Verification:** The high baseline performance confirms that the models possess the requisite lexical knowledge in their pretraining distribution. Therefore, the failures observed in the scrambled settings (Section 4.2) cannot be attributed to a lack of semantic understanding, but rather to a specific deficit in structural reconstruction.

2. **Metric Validation:** The high scores under canonical conditions validate our evaluation pipeline. They indicate that our automated metrics accurately reward correct semantic interpretations, ruling out the possibility that the low scores in the main experiments are artifacts of an overly punitive scoring mechanism.

The stark contrast between this high-performance baseline and the significant degradation observed under permutation (where recovery rates frequently drop below 35%) provides compelling evidence for the semantic–structure dissociation hypothesis discussed in the main text.

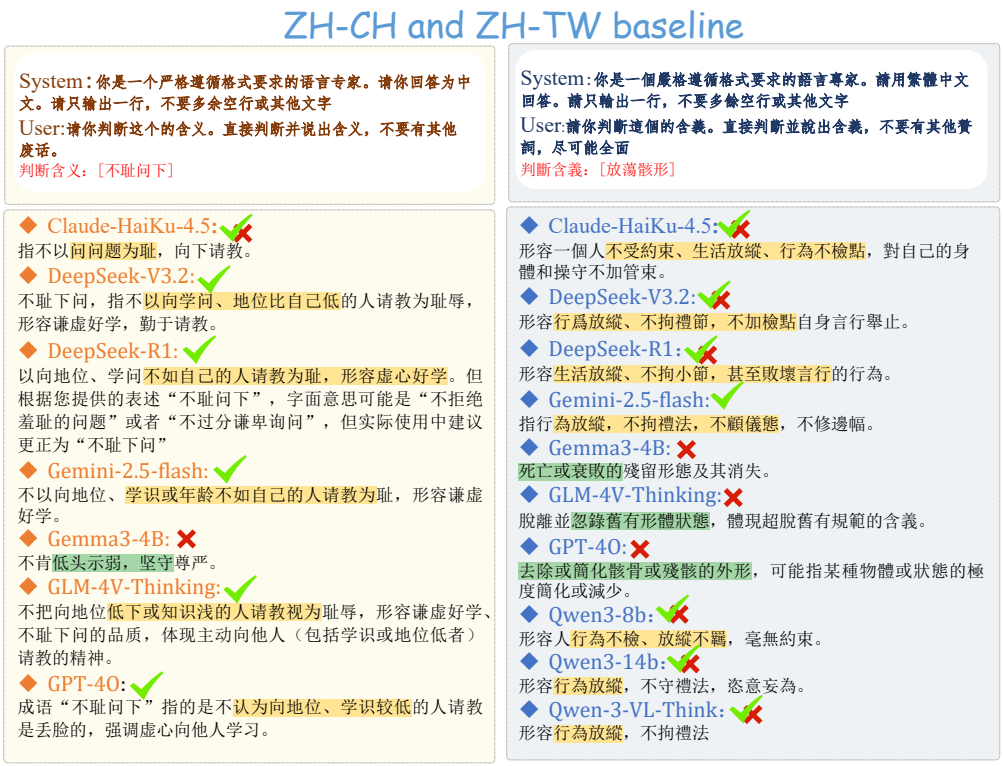

Figure 8: **Baseline performance on Chinese scripts illustrating Simplified Chinese on the left and Traditional Chinese on the right.** Even powerful models often fail to recover the canonical order.

Figure 9: **Baseline performance on Japanese and Korean scripts showing Korean Hangul on the left and Japanese Kanji on the right.** The structural scrambling effectively disrupts local semantic anchors.